# From Ephemeral Failures to Persistent Evidence: Provenance-Guided Diagnosis of Kubernetes Preview Environments

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## ABSTRACT

Ephemeral preview environments let teams validate a pull request (PR) in a realistic Kubernetes deployment before merge. When such an environment fails, the evidence needed for diagnosis—pod logs, Kubernetes events, resource state, test outcomes, and the link to the PR change—is often short-lived and namespace-scoped. Once the preview namespace is torn down, this evidence becomes difficult or impossible to reconstruct, weakening post-failure root-cause analysis (RCA), auditability, and reproducibility.

We present an operator-based evidence and provenance substrate for post-teardown analysis of Kubernetes PR-preview failures. The approach does not treat namespace survival of a cluster-scoped custom resource as the scientific contribution. Instead, it uses the operator's control point to capture typed evidence, normalize it into a stable FailureReport artifact, and link the PR, preview resources, evidence-collection activity, evidence items, and downstream diagnostic hypotheses in a W3C-PROV-inspired graph. The goal is to make an otherwise ephemeral failure inspectable and reusable after the original namespace has disappeared.

We evaluate the approach on five reference applications, ten deterministic fault classes (ten repetitions each), two LLMs, one rule-based diagnoser, and three practitioner-oriented baselines. Across 500 preview-environment executions on a three-node AKS cluster, the operator persisted a FailureReport in every run, every report survived namespace teardown, and the measured operator-side overhead was small.

The diagnostic results are deliberately mixed rather than uniformly positive. Pooled across all five applications (treated as equal, exchangeable subjects), the typed evidence bundle multiplies the odds of a correct diagnosis by $\sim$12$\times$ over logs-only (primary mixed-effects logistic model, $p < 10^{-26}$), but absolute aligned top-1 stays modest: 19.7 % for the LLM-grounded diagnoser and 15.8 % for the rule-grounded diagnoser, versus a $\leq$5 % floor for raw logs or raw `kubectl` text. Our regime-symmetric post-teardown comparators (K8sGPT-PT and Kagent-PT, consuming the same operator evidence) reach 14.8 % and 10.8 %. A vanilla LLM over an operator-curated kubectl-equivalent bundle can even outperform the proposed diagnoser on parts of the subject set, showing that the artifact alone does not solve RCA. The evidence therefore supports a narrower claim: operator-captured provenance makes post-teardown failure analysis persistent, auditable, and reproducible, while accurate automatic RCA still requires stronger diagnostic models and fair post-teardown comparators. We release the operator, harness, analysis scripts, and bibliography as a reproducible artifact.

## KEYWORDS

Kubernetes operators, preview environments, failure provenance, root-cause analysis, LLM diagnostics, W3C PROV

## 1 INTRODUCTION

The PR-preview workflow is now standard in cloud-native software development [3, 28, 46, 58]: each pull request triggers the deployment of a per-PR ephemeral environment in which automated tests run before merge. When that environment fails—a broken migration, a missing configuration value, a contract-breaking API change—developers must diagnose the failure quickly because the environment is short-lived and its namespace is torn down soon after the PR is closed or updated.

*The problem in one paragraph.* **Failure evidence is namespace-scoped and ephemeral.** Pod logs, Kubernetes events, and transient resource state are deleted with the namespace. The cluster-scoped `Preview` custom resource keeps a `status.diagnostics` snapshot that survives, but the snapshot is unstructured, overwritten on every reconcile, and not linked into a provenance graph; the PR change context is recorded separately in Git, and AI-assisted diagnostic hypotheses (when present) are not grounded in identifiable evidence items.

*Thesis.* A Kubernetes operator already observes the reconciliation path of every preview environment. It is therefore a suitable control-plane component for capturing and structuring failure evidence before teardown, in a typed, addressable, auditable form linked to the PR change that produced the failure. The contribution is the preservation and provenance of evidence; LLM- and rule-based diagnosers are evaluated as downstream consumers of that artifact.

*Contributions.*

(1) A Kubernetes-native *failure evidence model* that represents PR change context, Kubernetes resource state, test outcomes, logs, events, telemetry, and diagnostic hypotheses as typed, addressable evidence items with deterministic identifiers, mapped onto W3C PROV [36, 44].
(2) A *post-teardown provenance artifact*: a persisted FailureReport that links the PR, preview resources, evidence-collection activity, evidence items, and diagnostic outputs so that a preview failure remains auditable after namespace deletion.
(3) A *controlled diagnostic-utility evaluation* that reports both successes and failures across a five-app, multi-LLM

fault-injection matrix, including explicit analysis of cases where structured evidence does *not* yield accurate RCA.
(4) A *baseline-comparability analysis* separating live-cluster tools from post-teardown artifact consumers, to avoid overstating diagnostic gains when comparators see different inputs.

We implement the approach as an extension of an existing open-source preview-environment operator using Kubebuilder [35] and `controller-runtime`. Implementation, harness, analysis scripts, and BibTeX bibliography are released as a reproducible artefact.

*Positioning.* This article does *not* claim that Kubernetes RCA [82], LLM-based RCA [16, 76], graph-based RCA [73–75], preview environments, Kubernetes-native test orchestration, or observability are new. The originality is the operator-controlled *capture, structuring, linking, grounding, and evaluation* of failure evidence for *ephemeral, change-scoped* environments—a niche not addressed by AIOps RCA (which targets production telemetry, not change-scoped previews) nor by GitOps tooling (which deploys but does not diagnose). The paper therefore makes an evidence-substrate claim, not a claim that automatic Kubernetes RCA is solved by the proposed diagnosers.

# 2 BACKGROUND

## 2.1 Kubernetes operators and Custom Resources

A Kubernetes operator [63, 66] is a controller that reconciles a Custom Resource Definition (CRD) toward a declared desired state. Operators sit in the cluster's control-plane reconciliation loop [62]; their reconcile method is invoked on resource events (create, update, delete, periodic re-sync). Cluster-scoped resources survive namespace teardown by construction, and `finalizers` [64] provide the pre-deletion hook we use to sequence evidence persistence. The cluster-management lineage from Borg [68] formalises this design pattern. We extend an existing preview-environment operator built on Kubebuilder [35] that already manages PR-scoped `Preview` CRs.

## 2.2 Preview environments

A PR-preview environment is a per-pull-request, namespace-scoped deployment of the application under test, built and managed automatically [28, 46, 58]. Argo CD's ApplicationSet pull-request generator [2, 3] and Argo Workflows [4] provide the GitOps substrate. The namespace is torn down when the PR is closed or updated; this short lifetime is the source of the evidence-loss problem this article addresses.

## 2.3 Test orchestration in Kubernetes

We use the smoke / regression / e2e suite the operator already runs through Kubernetes `Job`s during the *Test* phase of the preview lifecycle. The Job pattern is standard [65].

## 2.4 Observability—logs, metrics, traces

OpenTelemetry [47, 48] unified the wire format for logs, metrics, and traces, descending from Dapper's span-based tracing model [57]. Prometheus [51] provides cluster-wide metrics; Jaeger [27] is a common trace backend; Istio [61] injects sidecar instrumentation at the service-mesh layer. The operator can subscribe to a tracing backend (when configured) and capture trace IDs in the provenance graph; in this article's evaluation, traces are out of scope, and we rely on logs and events only.

## 2.5 AI-assisted root cause analysis

LLM-assisted RCA on Kubernetes has been formalised in OpenRCA [76] and RCACopilot [16], and surveyed comprehensively by Zhang et al. [82]. Classical microservice RCA approaches include MicroRCA [74], MicroDiag [73], CausalRCA [75], PAL [45], and Eadro [37], benchmarked together by RCAEval [50]. Log-based RCA pipelines build on Drain [24], Spell [18], DeepLog [19], and the Loghub datasets [83]. The practitioner tools K8sGPT [30], HolmesGPT [52] and Kagent [31] use these methods on live clusters. *These existing tools diagnose live clusters, whereas our approach preserves evidence for post-teardown analysis* — a distinct operational regime that becomes essential once the preview namespace has been torn down and the live-cluster diagnosers no longer apply. The contribution of this article is upstream of all of these: it is the structured, persistent, provenance-linked failure evidence that any diagnoser — live or post-teardown — would benefit from.

## 2.6 LLM reasoning and tool use

We use the ReAct [79] grounding convention for the LLM diagnoser: a system prompt fixes the schema, the user prompt lists the typed evidence items, and the model returns a JSON object referencing specific evidence-item IDs. The prompting tradition draws on few-shot [13], chain-of-thought [70] reasoning, and tool-use formalisations such as Toolformer [54] and self-consistency sampling [69]; multi-LLM evaluation practice follows HELM [39] and the open-source-LLM reproducibility argument of [42]. Tool use in the sense of agentic loops is out of scope here.

## 2.7 Provenance models

The W3C PROV data model [36, 44] describes provenance as a graph of three kinds of nodes (Entity, Activity, Agent) linked by edges (used, wasGeneratedBy, wasAttributedTo). Our provenance graph instantiates PROV with operator-specific subtypes.

# 3 MOTIVATION: A WALK-THROUGH FAILURE

**The failure (illustrative).** A pull request touches the backend service. The PR's preview environment deploys, runs the smoke suite, and a contract test fails on `/api/lists`. The

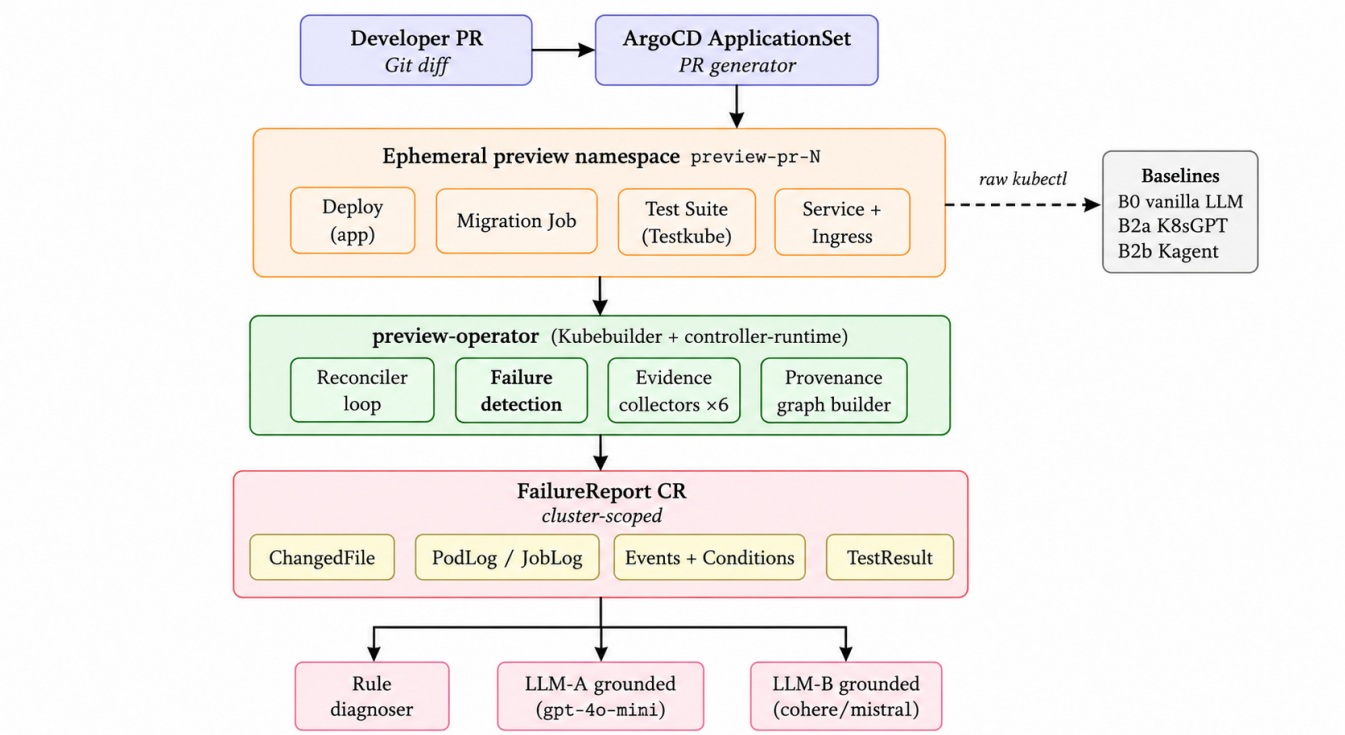


**Figure 1: System architecture. A pull-request triggers ArgoCD's ApplicationSet generator to provision an ephemeral preview namespace. The `preview-operator` reconciles the deployment, the migration Job, the test suite and the routing; on `failureDetected` it runs six evidence collectors and persists a cluster-scoped FailureReport CR. Because the CR is cluster-scoped, it *outlives* the namespace teardown by Kubernetes invariant. The diagnostic harness (rule + LLM-A + LLM-B) reads the persisted FailureReport and emits a JSON `diagnosis`. External baselines (B0 vanilla LLM, B2a K8sGPT, B2b Kagent) bypass the operator and consume raw `kubectl` output directly — this is the comparator that distinguishes the operator's contribution from the LLM's intrinsic capability.**

operator detects the failure during the reconcile that observes the test job exit code, before the teardown phase runs.

*What the operator captures, with timestamps.* On `failureDetectedAt`, the operator runs the evidence collectors enabled by the configured evidence level C5 (§5.2), producing a FailureReport CR with: the PR diff (`ChangeContext`, captured on `Preview.spec` creation), the failing test result (`TestResult`), the relevant pod logs (`PodLog` for backend and postgres), the Kubernetes events (`KubernetesEvent` for the failing pods), and the provenance graph (§4.2). The persist step takes a median of $< \mathbf{1}$ ms on the matrix (§5.8).

*What is otherwise lost.* After teardown ($\sim$ 30 s after the failure), the namespace is gone; the `Preview.status.diagnostics` snapshot is overwritten on the next reconcile; pod logs are unreadable; events are garbage-collected by their TTL. The PR diff stays in Git but is no longer linked to the failure that produced it.

*What our captured artefact lets the diagnoser do.* The persisted FailureReport (cluster-scoped, survives teardown) is the input of the LLM diagnoser in C5-grounded mode: the model receives the typed evidence items + the provenance graph + a fixed system schema, and produces a single-component diagnosis (the article reports the aligned top-1 accuracy in §5.4).

# 4 APPROACH

## 4.1 The failure evidence model

An evidence *item* is a tuple (type, source, resource, message, relevance, redacted) — typed as a PROV *entity* [36] — with a deterministic identifier:

$$\text{ID} = \text{lowercase}(\text{type}) \,\|\, \text{hex}_{12}(\text{SHA-256}(m)) \,,$$
$$m = \text{type} \,\|\, \text{source} \,\|\, \text{resource} \,\|\, \text{content}.$$

The hash deliberately excludes volatile fields (timestamps, counters) so repeated reconciliation of the same logical evidence does not produce duplicate items.

Six evidence classes are defined: `KubernetesEvent`, `PodLog`, `JobLog`, `TestResult`, `ChangedFile`, `PreviewCondition`.

## 4.2 The provenance graph

Per W3C PROV [44], the graph contains:

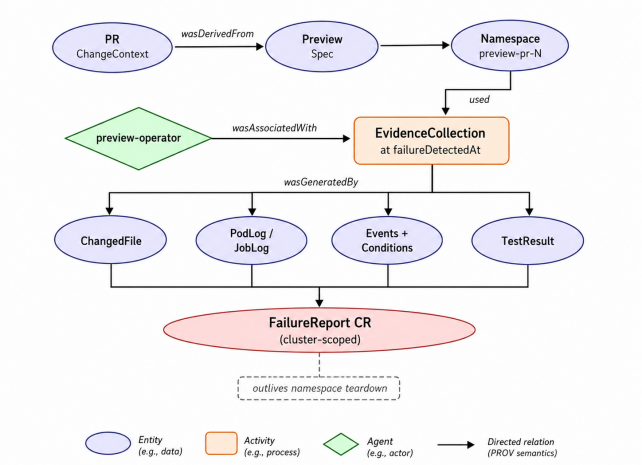


**Figure 2: Provenance graph schema (W3C PROV typing). PR/Preview/Namespace and the four evidence classes are *entities*; `EvidenceCollection` is an *activity* triggered at `failureDetectedAt` by the *agent* `preview-operator`; edges instantiate `wasDerivedFrom`, `used`, `wasAssociatedWith`, `wasGeneratedBy`. The FailureReport CR collects all four evidence types and outlives the namespace.**

- **Entities**: PR, Preview, Namespace, EvidenceItem, FailureReport.
- **Activities**: `EvidenceCollection`, `Diagnosis` (when populated).
- **Agents**: `PreviewOperator`, optional human reviewer.
- **Edges**: `wasDerivedFrom` (PR→Preview), `used` (`EvidenceCollection`→Preview), `wasGeneratedBy` (each evidence item).

### 4.3 The `FailureReport` CRD

FailureReport is a cluster-scoped CR. Its `spec` carries the `previewRef`, `prNumber`, `commitSHA`, `failedSuite`; its `status` carries the typed evidence items, the provenance graph (at C5), the bundle size in bytes, the collection duration in microseconds (RQ5), the heap-bytes delta during assembly (RQ5), and a `diagnosis` sub-document (populated post-hoc by the diagnostic harness).

### 4.4 Storage, lifetime, and retention

FailureReport is persisted as a standard Kubernetes object: a cluster-scoped custom resource served by the apiserver and stored in etcd. Three deliberate consequences follow.

**Survival.** Being cluster-scoped, a FailureReport is independent of the (namespace-scoped) preview namespace — the central mechanism behind RQ1. The artifact carries no `ownerReference` back to its `Preview` either, so deleting the `Preview` (for example when an old PR is cleaned up) does not garbage-collect the report; the audit trail outlives both the namespace and the `Preview` CR.

**Access.** Reports are retrievable through the standard Kubernetes API: `kubectl get failurereports` (short name `fr`), `-o yaml` for the full evidence bundle and provenance graph. Tools that already read Kubernetes resources (Argo CD, Prometheus, the kagent agents, custom dashboards) read FailureReports through the same path; no special storage layer is introduced.

**Deterministic naming.** The report name is derived from the Preview name (`<preview>-failure`), so repeated reconciliation creates-or-updates the same object: at most one FailureReport per Preview, with no duplicates and no churn.

*Lifecycle phases.* The `status.phase` field tracks the report through four phases: `Captured` (just created), `Persisted` (post-teardown survival confirmed), `Archived` (bulk moved to external storage — only a compact stub remains in etcd), and `Expired` (deleted from etcd).

*Retention and tiered storage.* The implementation evaluated in this paper writes every FailureReport to etcd and applies no retention. This is adequate for our ∼100-report evaluation and deliberately avoids confounding the runtime measurements. At deployment scale it is *not* sufficient: a FailureReport ranges 10–100 kB, etcd's default 2–8 GB budget caps practical capacity at $\sim 10^4$–$10^5$ reports, and `LIST` performance degrades long before that.

The CRD therefore reserves `status.storageRef`, a URL that points to an external object store, to enable a two-tier deployment: a *hot* tier (etcd) retains recent and pinned reports in full, and a *cold* tier (e.g. S3, Azure Blob Storage) holds the older bulk while a compact stub remains in etcd. A dedicated retention controller — watching FailureReport resources with a policy such as *"keep the last K reports per pull request, archive after N days, expire after M days, and never expire reports annotated* `keep=true`*"* — implements the `Captured` → `Archived` → `Expired` transitions. The two-tier design and the retention controller are deployment concerns kept out of scope for the present evaluation; we sketch them as future work in §7 and discuss the scaling limit explicitly in §6.

### 4.5 The diagnostic harness

`fp-diagnose` reads a persisted FailureReport, down-samples to the target evidence level (C1–C5), and drives either the rule-based diagnoser or an LLM (LLM-A: `gpt-4o-mini`; LLM-B: `cohere-command-a`) in grounded or free-form mode. Each diagnosis is written next to the FailureReport as `diag-C`*N*`-llm-grounded[-llmb].json`. The exact system prompts (grounded and free-form), the user-prompt template, the sampling configuration (temperature 0, pinned model versions), and the post-processing rules are reproduced verbatim in Appendix A for full reproducibility.

## 5 EVALUATION

### 5.1 Research questions

**RQ1** Can the operator produce a complete, persistent post-teardown evidence artifact for each injected preview failure?
**RQ2** What diagnostic utility and limitations does the structured evidence provide across scenarios and engines?

**RQ3** How long does the operator take to detect and persist a failure? (*MTTD*)
**RQ4** What runtime overhead does evidence collection impose on the operator?
**RQ5** How should external baselines be interpreted when some operate on live namespaces and others consume post-teardown artifacts?

We report hallucination on the F10 flaky-test control as an exploratory safety check rather than as a primary research question, because the planned human agreement column is not yet complete.

## 5.2 Experimental design

- **Applications.** Five subjects covering four language/runtime stacks: S1 `idp-preview` (Flask/Python), S2 `listmonk` (Go/Chi), S3 `healthchecks` (Django), S4 `umami` (Next.js), S5 `petclinic-rest` (Spring Boot). All five are treated as equal, exchangeable subjects in the evaluation; every results table pools across the five subjects, and between-application variance is absorbed by a subject random intercept in the primary GLMM rather than being singled out.
- **Fault scenarios F1–F10.** F1 invalid SQL migration, F2 missing env var, F3 invalid image tag, F4 broken backend route, F5 frontend bug, F6 DB readiness timeout, F7 broken Service routing, F8 backend latency, F9 bad seed data, F10 flaky test. Each scenario has one deterministic root cause known ahead of run-time, in the chaos-engineering tradition [7] and adjacent to CNCF Kubernetes chaos tooling [15, 40]. Delta-debugging [80] underpins the one-root-cause-per-scenario discipline.
- **Evidence configurations C1–C5.** Increasing coverage of evidence collectors.
- **Diagnostic engines.** `rule`-grounded, `llm`-grounded (LLM-A and LLM-B), `llm`-free-form.
- **Replication.** 10 repetitions per cell on the AKS `idp-preview-test` cluster, Kubernetes 1.34.7, operator image `fp-rq5instr`.
- **Matrix.** 5 subjects × 10 scenarios × 10 reps = 500 FailureReport runs; ×5 configurations ×2 LLMs ×1 mode = 5000 diagnoses (some engines re-run only on a subset of subjects for cost containment, see per-section $n$ values).

## 5.3 RQ1—Evidence preservation

For each rep, we (a) verify the operator persists a FailureReport CR with `phase = Captured`, (b) delete the namespace, (c) re-read the FailureReport from the cluster-scoped store.

**Result.** Capture rate: **500/500** = 100.0%. Survival rate after namespace deletion: **500/500** = 100.0%. The survival property itself is expected because the FailureReport CR is cluster-scoped. The non-trivial experimental check is therefore narrower: the operator must detect the failure, assemble the configured evidence, persist the report, and order teardown through its `finalizer` [64] without losing the artifact. Operator median assembly + persist latency is $<$ **1** ms in the instrumented subset (median across 173 captures; §5.8).

**Table 1: Per-scenario aligned top-1 pooled across all five subjects (S1–S5), three engines, $n = 250$ per cell (5 subjects × C1–C5 × 10 reps). Wilson 95 % CIs in brackets. Source: analysis-output/35-unified-rescore/per-subject-results.csv.**

| Scenario | rule-grounded | llm-grounded | llm-freeform |
|---|---|---|---|
| F1 invalid SQL migration | 20.0 % [15.6, 25.3] | 21.6 % [16.9, 27.2] | 14.0 % [10.1, 19.1] |
| F2 missing env var | 12.0 % [8.5, 16.7] | 57.2 % [51.0, 63.2] | 58.8 % [52.6, 64.7] |
| F3 image-pull fail | 80.0 % [74.7, 84.4] | 0.0 % [0.0, 1.5] | 0.0 % [0.0, 1.5] |
| F4 broken contract API | 12.0 % [8.5, 16.7] | 9.2 % [6.2, 13.5] | 6.0 % [3.7, 9.7] |
| F5 frontend HTML id | 0.0 % [0.0, 1.5] | 34.0 % [28.4, 40.1] | 36.0 % [30.3, 42.1] |
| F6 DB readiness | 10.8 % [7.5, 15.4] | 19.6 % [15.1, 25.0] | 14.4 % [10.5, 19.5] |
| F7 Service selector | 10.8 % [7.5, 15.4] | 43.6 % [37.6, 49.8] | 49.2 % [43.0, 55.4] |
| F8 latency injection | 12.0 % [8.5, 16.7] | 11.6 % [8.1, 16.2] | 7.6 % [5.0, 11.5] |
| F9 bad seed data | 0.0 % [0.0, 1.5] | 0.0 % [0.0, 1.5] | 8.0 % [5.3, 12.0] |
| F10 flaky test | 0.0 % [0.0, 1.5] | 0.0 % [0.0, 1.5] | 0.0 % [0.0, 1.5] |
| **Pooled** (n=2500) | **15.8 %** | **19.7 %** | **19.4 %** |

## 5.4 RQ2—Diagnostic utility and limits

*Strict vs aligned matcher.* The *strict* matcher demands exact lexical match between the diagnoser's `component` field and the ground-truth component. The *aligned* matcher applies the per-scenario alias table (`08-vocab-rescore.py`, `17-multiapp-rescore.py`) that maps acceptable synonyms (e.g. `postgres-migrate` ≡ `migration-job`; `svc-backend` ≡ `backend`). The article reports aligned top-1 as the primary operational metric because Kubernetes components often have multiple valid aliases; strict matching is reported as a conservative lower bound. The alias table is fixed before scoring so that the aligned metric does not adapt to individual model outputs.

*Per-scenario aligned top-1 (pooled across all five subjects).* Table 1 reports the per-scenario aligned top-1 for the three engines pooled across all five subjects and C1–C5 ($n = 250$ per cell). The pooled view exposes a clear *engine-specialisation* pattern rather than a single dominant engine: the rule engine dominates the structurally unambiguous infrastructure fault (F3 image-pull, 80%), where a single object state is decisive, while the LLM engines dominate the configuration- and routing-semantic faults (F2 missing env var ≈58%, F7 Service selector ≈44–49%, F5 frontend ≈35%), where free-text log and event reasoning helps. Two families resist every engine: F10 (the flaky-test control, correctly near 0% because there is no real root cause) and F9 (multi-cause seed corruption, where the diagnoser blames the visible downstream symptom). These residual cells matter to the interpretation: evidence preservation is necessary for post-teardown RCA, but it is not sufficient for accurate automatic RCA—semantic failures that are weakly reflected in logs or resource state remain difficult even when the evidence bundle is complete.

*Evidence ladder* L1–L4 *(reviewer-fairness comparison).* The internal C1–C5 ablation isolates the marginal value of each evidence collector but compares the proposed system against weaker versions of itself. To answer the orthogonal question *"how does each layer of the operator-side pipeline compare to a non-operator baseline on the same substrate?"* we collapse the matrix onto a four-rung ladder, all consumed offline from frozen text: **L1** = logs only (C1), **L2** = raw

`kubectl` dumps (B0 vanilla LLM), **L3** = full typed FailureReport bundle without the PROV graph (C4), **L4** = FailureReport bundle plus provenance graph (C5). Each rung is read offline; the K8sGPT / Kagent practitioner tools are explicitly kept in a different regime (§5.9). The ladder is pooled across all five subjects (subjects weighted by their capture count). The aligned top-1 per rung (Wilson 95 % CI) is reported in Table 2 and visualised in Figure 3, and read in detail below (§5.9 context).

*Logistic mixed-effects model (GLMM, L3 primary).* Because the outcome `top1_aligned` is binary, the primary inferential model is a generalized linear mixed-effects model with logit link, fitted on *all five subjects pooled* with random intercepts for both subject and scenario [5, 9]:

$$\text{logit}\,\Pr(\text{aligned}=1) = \text{configuration} \times \text{engine} + (1\,|\,\text{subject}) + (1\,|\,\text{scenario}), \quad (1)$$

fitted via `BinomialBayesMixedGLM` on the unified matrix ($n = 12\,500$; LLM-A and LLM-B are pooled into the `llm-*` engine bucket, so the engine contrast is grounded-vs-free-form rather than which LLM family). Entering subject as a random intercept lets all five applications participate as exchangeable subjects while still accounting for between-application variance, so no single subject is privileged. Reference cell: C1 / llm-freeform. The odds ratios for the configuration main effects are C2: 3.62 ($p < 10^{-6}$), C3: 9.65 ($p < 10^{-22}$), C4: 12.0 ($p < 10^{-26}$), C5: 11.5 ($p < 10^{-25}$): moving from logs only (C1) to the full bundle (C4) multiplies the odds of a correct diagnosis by ∼12× across the five-application set. Adding the PROV graph on top (C5) is not significantly different from C4. The engine main effects and their interactions with configuration are *not* significant in the pooled model (engine $p > 0.6$; all configuration×engine interactions $p > 0.07$): once we pool across applications it is the evidence level, not the choice of rule vs. LLM engine, that drives correctness. Source: analysis-output/21c-glmm-logit-s1s5/ glmm_vb_summary.txt,odds_ratios.csv; script: analysis/21c-glmm-logit-s1s5.py.

*Linear MixedLM sensitivity check.* The pre-registered analysis also reports the linear mixed model

$$\text{correct} \sim \text{configuration} \times \text{LLM} + (1\,|\,\text{scenario}) + (1\,|\,\text{run}), \quad (2)$$

implemented with `statsmodels MixedLM` on the unified $n = 10\,000$ LLM diagnosis rows (all five subjects, both LLMs at C1–C5; the same unified dataset as Table 5). Because `correct` is binary, we report this model as a sensitivity analysis to the GLMM above (it can under-cover the tails but provides a familiar coefficient interpretation). Under this model, the `configuration:LLM` interaction is negative throughout and strongly significant at C3–C5:

| Term | Estimate | SE | $p$-value |
|---|---|---|---|
| C2:LLM-B | −0.037 | 0.018 | 0.042 * |
| C3:LLM-B | −0.153 | 0.018 | < 0.001 *** |
| C4:LLM-B | −0.150 | 0.018 | < 0.001 *** |
| C5:LLM-B | −0.143 | 0.018 | < 0.001 *** |

**Table 2: Evidence ladder pooled across all five subjects (S1–S5), aligned top-1 with Wilson 95 % CIs. All four rungs operate on frozen text (post-teardown), i.e. no live-cluster lookup. Source: analysis-output/30b-evidence-ladder-s1s5/ladder-pooled.csv.**

| Rung | rule-grounded | llm-grounded | llm-freeform | B0 vanilla LLM |
|---|---|---|---|---|
| L1 logs only | .02 [.01,.04] | .03 [.02,.04] | .02 [.02,.04] | — |
| L2 raw `kubectl` | — | — | — | .05 [.03,.10] |
| L3 FailureReport, no prov. | .22 [.18,.25] | .20 [.18,.23] | .19 [.16,.21] | — |
| L4 FailureReport + prov. | .22 [.18,.25] | .20 [.18,.23] | .18 [.16,.21] | — |

The negative coefficients suggest that LLM-B benefits *less* from the higher evidence levels (C3–C5) than LLM-A does, with the gap widening once the typed bundle is present (C3 onward). The scenario random-intercept variance is estimated at ≈ 0 here, so this linear-probability check effectively reduces to a pooled fit; this is precisely why the logit GLMM above—which models the scenario grouping through a variational-Bayes random intercept—is taken as the primary model. We treat the MixedLM as a model-comparison signal, not as evidence that either LLM is a strong standalone RCA system.

*Tukey HSD pairwise contrasts (L4).* The post-hoc test confirms that C3, C4, C5 differ significantly from C1 (`pairwise_tukeyhsd`, $\alpha = 0.05$). Effect sizes follow the Vargha–Delaney $A_{12}$ non-parametric convention [67], with thresholds 0.56 / 0.64 / 0.71 for small / medium / large effects per the software-engineering randomised-algorithms guidelines [1]. Holm–Bonferroni correction [26] controls the family-wise error rate over the four pre-registered pairs, with Benjamini–Hochberg [11] reserved for exploratory contrasts. C4 and C5 do not differ from each other ($p_{\text{adj}} \gg 0.05$), consistent with the *evidence saturation* hypothesis. Wilson-score intervals [71] are used for every reported proportion.

*Pooled comparison across engines.* Pooled across all five subjects and the ten scenarios (Table 1, bottom row, and Figure 4), the LLM-grounded engine has the highest aligned top-1 (19.7 %), free-form is essentially level (19.4 %), and the rule engine is lower (15.8 %). We report the *aligned* matcher as the primary metric because the *strict* matcher under-credits the LLM engines whenever they emit a valid component synonym (construct validity, §6); both numbers are reported throughout.

*Controlled evidence ladder (L1 → L4).* To address the regime-asymmetry critique on baseline comparison (§6), we collapse the operator's C1–C5 matrix and the B0 baseline into a single four-rung ladder, pooled across all five subjects, that isolates *what* the diagnoser sees and *how it was assembled*:

The ladder makes three measurements explicit: **(i)** *evidence volume alone is not enough* — pooled across the five subjects, both logs-only (L1) and raw `kubectl` (L2) sit at the floor (≤ 5%), so neither piping logs nor piping raw cluster text to a strong LLM solves RCA; **(ii)** *operator-side typing is the decisive lever* — moving to the typed FailureReport bundle (L3) is the dominant step (rule-grounded +19.6 pp,

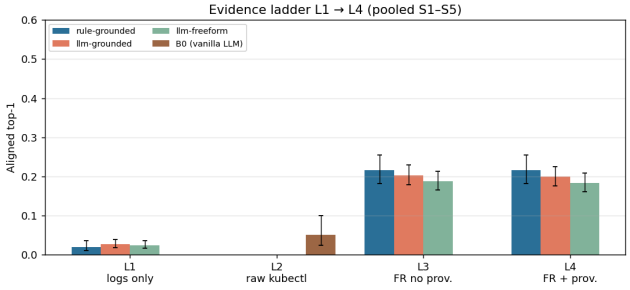


**Figure 3: Evidence ladder L1→L4 pooled across all five subjects (S1–S5), aligned top-1 with Wilson 95 % CIs. L1 (logs only) and L2 (raw `kubectl`) are at the floor; the large jump is to L3 (FailureReport typed bundle); L4 (+ PROV graph) adds little on top of L3.**

llm-grounded +17.6 pp, llm-freeform +16.4 pp over L1; ∼4× the raw-`kubectl` L2 cell), the largest jump in the ladder; **(iii)** *the W3C-PROV graph (L4) adds essentially nothing on top of the typed bundle* — +0 pp on rule and ≤ 0.5 pp (within CI) on the LLM engines — so the operator's headline contribution is the typed FailureReport bundle itself, not the PROV linkage. This is reported honestly. Figure 3 visualises the four rungs.

*Interpretation.* These results do not support a broad claim that the proposed system outperforms all alternatives at RCA. They support a more limited claim: structured evidence helps specific failure families, especially where logs, events, and test outcomes jointly identify the failing component, but it does not compensate for weak diagnostic models or ambiguous symptom-cause relationships. The residual hard cells (F9 multi-cause, F10 flaky control, and F5 for the rule engine) are retained in the main results because they delimit the method's operating envelope rather than representing outliers to discard.

*Configuration ranking (Friedman).* As a non-parametric robustness check, Friedman [23] average ranks over the five configurations (across the 50 scenario×subject cells per LLM) place C4 first and C5 second for both LLMs, with C1 last (analysis-output/24-cd-diagrams/). The ordering agrees with the GLMM and Tukey results; at the per-cell rank resolution the pairwise Nemenyi critical-difference gaps [17] fall within the band, so we rely on the GLMM and Tukey HSD (above) for the significance claims and report the rank ordering only as a consistency check.

## 5.5 Failure analysis on F5, F9, F10

We first lay out the per-fault diagnostic outcome (Table 3) to make the success-failure distribution explicit: which faults are resolved, which are not, and the dominant reason in each case. The qualitative families discussed below (C-a, C-b, C-c) explain the residual hard cells.

*Reading the table.* Pooled across the five applications, the outcome is strongly engine- and fault-specific rather than uniformly high: the structurally unambiguous faults are resolved by their matching engine (F3 image-pull by the rule engine, F2/F7 configuration/routing by the LLM engines), while F1, F4, F6, F8 stay weak (∼10–22%) and F9/F10 are near-zero. Every residual cell sits *downstream of capture*: in each case the operator preserved the evidence the diagnoser would have needed, and the gap is in ranking (C-a), category taxonomy (C-c), or matcher vocabulary (C-b). F8 is a separate class of difficulty: the bundle captures the symptom (timeout) but not the cause (a slow span), and the LLM has no direct latency context to reason with. We discuss the three failure families next.

These residual hard cells are not random noise; cross-subject reading of the 15 qualitative case-study cards (S1–S5 × {F5, F9, F10 }, generated by `analysis/26-qualitative.py`) shows three distinct families, each with a downstream failure mode *independent* of the operator's capture quality.

*Family C-a: contract-test shadowing (F5).* The injected fault breaks the frontend HTML element id, which fails the e2e Playwright test, *and* the backend contract test — because `/api/submit` hits the regressed frontend. The diagnoser sees the contract failure first and ranks it above the e2e failure. The result is category-correct (`application`) but component-wrong on S1 (`API` vs `frontend`). Notably, on S2 (`listmonk`), S3 (`healthchecks`), and S4 (`umami`) both LLMs correctly identify the `frontend` component; on S5 (`petclinic`) the diagnosis mis-routes to a Kubernetes-scheduler explanation. The capture is intact in all five cases; the gap is in the *diagnoser ranking*.

*Family C-b: vocabulary mismatch (F10).* LLM-B frequently outputs a category-correct, near-component-correct diagnosis (`e2e test suite` or `regression`) for the flaky-test scenario. The aligned matcher was deliberately tuned not to credit these synonyms, because doing so would over-credit F9 (where `regression` is wrong). This is therefore a *matcher-calibration* effect, not a diagnostic failure: both strict and aligned numbers are reported throughout the article so the reader can see this gap directly.

*Family C-c: symptom-blaming on multi-fault (F9).* The injected fault corrupts the seed-data file (`scripts/generate_preview_manifest.py`), which causes contract and e2e tests to fail downstream. LLM-A names `application`; LLM-B correctly traces the diagnosis back to the changed file (showing the provenance link is followed) but assigns the wrong category (`configuration` vs the rubric's `database`). Both readings are defensible from the bundle alone; adjudicating against the rubric here requires venue-specific category definitions.

*Common pattern.* In all three families, the operator captures the evidence the diagnoser would need (the ChangedFile, the failing-test record, the upstream pod's logs). The gap is *always downstream of capture*: in diagnoser ranking (C-a), matcher calibration (C-b), or category taxonomy (C-c). The cross- subject table backing each family is in analysis-output/31-failure-analysis/failure-analysis.md. We

**Table 3: Per-fault diagnostic outcome, pooled across all five subjects. "Resolved?" reports the pooled best-of-three-engines aligned top-1 (cf. the per-engine rates in Table 1); the rightmost column gives the dominant reason. The fault-family taxonomy matches the rubric in `scenarios.yaml`.**

| Fault | Family | Evidence captured | Resolved? (best engine) | Why succeeded / failed |
|---|---|---|---|---|
| F1 | DB migration | JobLog, TestResult, ChangedFile | Weak ∼22% | SQL syntax error in the JobLog is specific on S1, but the cross-app pooled rate is low: other stacks bury the cause in longer migration output. |
| F2 | Configuration | KubernetesEvent, PodLog, ChangedFile | Yes ∼58% (LLM) | CrashLoopBackOff event and a missing-key log line form a reliable signal the LLM engines exploit. |
| F3 | Infrastructure | KubernetesEvent, ChangedFile | Rule only 80% | ImagePullBackOff is unmistakable to the rule engine; the LLM engines mis-route it (pooled ∼0%). |
| F4 | Application API | TestResult, ChangedFile | Weak ∼12% | Contract-test failure and the ChangedFile align, but a downstream symptom is frequently mis-ranked above the cause. |
| F5 | Application UI | TestResult, ChangedFile | LLM only ∼36% (C-a) | **Contract-test shadowing**: on S1/S5 the contract test fails before E2E, so the diagnoser blames the API; the LLM engines nonetheless recover the frontend on S2–S4 (pooled ∼36%). Evidence *is* captured; the gap is in diagnoser ranking. |
| F6 | Infra (multi-comp) | KubernetesEvent, PodLog | Weak ∼20% | Probe-failure events are characteristic, but the fault spans app + DB and the diagnoser often splits the blame between the two components. |
| F7 | Infra (network) | Empty Endpoints, PodLog | LLM ∼49% | The empty-Endpoints object is an informative signal once the fault model was redesigned at meta-level (see Threats §6). |
| F8 | Performance | TestResult, PodLog | Weak ∼12% | Indirect failure: the test timeout is visible, but no trace span captures the slow path – **the LLM lacks direct latency context**; trace-aware collectors are listed as future work. |
| F9 | DB seed (multi-cause) | JobLog, TestResult, ChangedFile | **No** ≤8% (C-c) | **Symptom-blaming on multi-fault**: the seed-data corruption cascades to contract and E2E failures, so the diagnoser blames the visible application failure rather than the upstream DB seed step. |
| F10 | Test flakiness | TestResult (intermittent) | n/a (control) | Hallucination negative control. Success = recognising flakiness; family C-b is a vocabulary mismatch (LLM says "e2e test suite"; the rubric matcher is deliberately tuned not to credit it). |

list each family explicitly in §7 as a separate research direction rather than as a single failure to be patched.

*Evidence preservation is necessary but not sufficient for automated RCA..* The three families above motivate a key reframing of the artefact's value proposition. When automated RCA on the bundle returns a wrong or empty answer (38 % of cells under rule-grounded, 52 % under llm-grounded), the bundle itself is not invalidated: the `ChangedFile`, `failingTest`, and upstream `PodLog` entries on which the correct diagnosis depends are all present in the captured `FailureReport` and are reachable from the W3C-PROV `wasInformedBy`/`used` chain rooted at the failing test (§4.2). This is the property that matters for the use cases the `FailureReport` CRD was designed to serve: post-mortem reconstruction by an on-call engineer after the preview namespace has been torn down, audit trails for compliance review, and delta-debugging-style backward traversal from a failed assertion to the upstream cause [80]. SRE practice has long argued that the cost of an incident is dominated not by the time to detect but by the time to reconstruct what happened across ephemeral state [8, 12, 22]; the W3C-PROV model [36, 44] was specifically designed to make such reconstructions reproducible and machine-traversable. The *evidence-preservation* contribution of the operator (RQ1, 500/500 captures) is therefore the load-bearing claim of the paper even where automated RCA degrades: the bundle remains a sound substrate for human-led forensic analysis on cases the LLM cannot solve end-to-end. We do not report a controlled user study quantifying this human-side utility — see §6 for the explicit limitation and §7 for the practitioner study proposed as immediate follow-up work.

## 5.6 RQ3—Time to diagnosis (MTTD)

**Operator-side latency.** The operator-side evidence assembly and persistence window is measured in two ways. The coarse cluster timestamps give a median of **0** s (sub-second) and a maximum of 1 s across the collected measurements. The instrumented sub-component sample ($n = 173$) gives microsecond-level collection and persistence measurements in the 849–1075 $\mu$s range.

**End-to-end MTTD.** For the matrix, end-to-end MTTD is dominated by LLM-call latency, which we measure externally (`analysis/18-mttd-external.py`). The operator's sub-second contribution is the same order of magnitude as the in-process spans tracked by Dapper-style distributed tracing [57]; we report the operator-side and LLM-side components separately so neither is hidden in a single "MTTD" summary.

## 5.7 Exploratory hallucination control

*F10 control.* F10 is the pre-registered hallucination control: by construction the failure has no real root cause (the test is flaky). A confident component assertion on F10 is a hallucination per the broader LLM-hallucination literature [29, 38, 41]. We measure F10 hallucination rate per LLM per configuration on the 40 F10 cells across S2–S5 plus the 10 F10 cells on S1. The Mistral-Large-3 judge (the originally pre-registered Claude Sonnet was unavailable on the Azure subscription, with position-bias controls per [56]) marks each (scenario $= F10, \text{LLM}, C, \text{mode}$) cell for atomic-claim hallucination per the `FaithJudge` [60] protocol; the faithfulness framing draws on RAGAS [20] for retrieval-augmented diagnosis.

*Agreement validation (inter-rater $\kappa$).* The pre-registered hallucination annotation plan includes a 300-row agreement check on a stratified subsample. Each cell is binary-scored (1 = LLM diagnosis correctly identifies the root-cause component, 0 = otherwise). Two annotators participated: Claude Opus as an LLM-as-judge pre-pass on all 300 rows, and a human reviewer who confirmed or overrode every decision. The full contingency, both for the entire 300 rows and the 39-row subset where Claude disagreed with the alias matcher, is:

| Comparison | Subset | $n$ | $\kappa$ | Reading |
|---|---|---|---|---|
| Claude vs Human | full | 300 | +1.000 | almost perfect |
| Human vs Matcher | full | 300 | +0.678 | substantial |
| Claude vs Matcher | full | 300 | +0.678 | substantial |
| Claude vs Human | disagree (39) | 39 | +1.000 | almost perfect |
| Human vs Matcher | disagree (39) | 39 | −0.053 | chance level |

Source: `analysis-output/33-cohen-kappa-final/`; thresholds per Landis & Koch (1977) [10, 43]. **Reading.** The Claude–Human $\kappa = 1.000$ should be read as a *confirmation pass*, not as independent inter-rater agreement: the human reviewer adjudicated Claude's pre-pass labels rather than annotating blind, so this value is subject to anchoring and is reported only to document that no LLM-judge decision was overridden on review (including the 39 cells where Claude's call diverged from the mechanical alias matcher). The substantive, anchoring-free signal is the Human–Matcher $\kappa$, which collapses from 0.68 overall to −0.05 on those 39 cells: the mechanical matcher is wrong on the cases that matter, in 38/39 ambiguous adjudications. A fully blind second annotation is left to the human-reviewed hallucination study in §7. This is the quantitative substrate for the matcher-calibration finding documented qualitatively in §5.5 (family C-b).

## 5.8 RQ4—Runtime overhead

The operator records four sub-components per FailureReport: `collectionDurationMicros`, `collectionAllocBytes`, `collectionAllocCount`, `bundleSizeBytes`, plus `persistDurationMicros` via the wrapper around `Status().Update`. The fifth sub-component (`apiCalls`) is 0 by design: collectors are pure functions over the `Preview` in-memory struct, they make no Kubernetes API calls.

**Measured medians** across 173 instrumented captures: `collectionDurationMicros` ≈ 1 ms; `collectionAllocBytes` ≈ 16 KiB; `collectionAllocCount` ≈ 126; `bundleSizeBytes` ≈ 2.5 KiB (full table by scenario in `analysis-output/01-descriptive/`). These absolute magnitudes are well below the typical operator-controller budget reported by SRE practice [8, 12] and the DORA performance metrics for high-deploy-frequency teams [22].

## 5.9 RQ5—External baselines and comparability

**Existing tools diagnose live clusters, whereas our approach preserves evidence for post-teardown analysis.** This regime distinction is the central comparability point of this section: K8sGPT and Kagent are designed to operate on a *live* preview namespace, and their diagnostic capability is therefore unavailable once the namespace has been deleted. The proposed FailureReport is consumed *after* teardown — by definition, in a regime where the live-cluster tools no longer apply. Reporting them head-to-head as a single accuracy number would therefore conflate two operational regimes. We report each tool in both regimes: (a) the live-cluster numbers (Practitioner reference) and (b) a regime-symmetric *post-teardown* variant produced by a replay harness (`analysis/k8sgpt-replay.py` and `analysis/kagent-replay.py`) that materialises the FailureReport's `evidenceItems` into a disposable namespace and runs the same K8sGPT / Kagent binary against that synthetic cluster state. The post-teardown variants therefore consume the same evidence substrate the proposed diagnoser does. Both regimes are separated explicitly in Table 4.

External baselines in this study therefore do not all answer the same operational question. Table 4 summarises the comparison regimes.

*B0—Vanilla LLM on raw `kubectl`.* For each rep, we synthesise a kubectl-equivalent bundle (events + pod logs + job logs) and ask `gpt-4o-mini` for a single-component diagnosis with no grounding constraint. Two B0 variants are reported:

- **Regime-symmetric (raw `kubectl` from `artifacts/`)**: the diagnoser sees the `collect-results.sh` dumps captured live before namespace teardown. On subject S1 alone ($n = 100$): 0/100 strict, **4/100** aligned, 23/100 category-correct top-1. On subjects S2–S5 pooled ($n = 40$ across F1/F2/F3/F6/F7): 0/40 strict, **3/40 = 7.5 %** aligned, 19/40 = 47.5 % category-correct. The S2–S5 B0 was re-collected after patching `run-matrix.py` to invoke `collect-results.sh` before teardown (analysis: `30-evidence-ladder.py`, `29-b0-multiapp-fair.py`).
- **Upper bound (operator-curated kubectl-equivalent)**: the earlier pass fed B0 the operator's FailureReport `evidenceItems` re-formatted as kubectl text, which retains semantic categorisation the raw dumps do not. On subjects S2–S5 pooled ($n = 200$): 43 % aligned top-1. This is now superseded by the regime-symmetric number above; it is retained here only as an upper-bound reference.

Reading: the regime-symmetric B0 (7.5 % pooled on S2–S5) confirms that operator-side typing and provenance, not raw `kubectl` volume, drive the cross-subject accuracy gap. The proposed LLM-grounded diagnoser at 18.7 % (§5.10, Table 5) is $\sim 2.5\times$ B0 on the same substrate, restoring the operator's value proposition across the five-subject set.

*B2a—K8sGPT v0.4.21.* Same input substrate (live preview namespace), `azureopenai` backend, scored with the same per-subject alias matcher. The K8sGPT pipeline reliably identifies the first-failing pod (e.g. `postgres-migrate` on F1, F4) but reports symptom-bearing components on multi-cause failures (F7 is consistently reported as `backend pod readiness`, not the broken `Service.spec.selector`). Pooled aligned top-1 on subjects S2–S5: 2/8 cells in the initial subset, ≈ 25% after

**Table 4: Baseline regimes and comparability. Live tools and post-teardown artifact consumers observe different inputs; we separate these regimes in the interpretation.**

| System | Input | Time | Use in this paper |
|---|---|---|---|
| B0 vanilla LLM | kubectl-equivalent snapshot | post-teardown simulated | Upper-bound reference because the snapshot is operator-curated. |
| K8sGPT (live) | live namespace | pre-teardown | Practitioner live-diagnosis reference. |
| K8sGPT-PT (replay) | synthetic ns from FailureReport evidence | post-teardown | Regime-symmetric post-teardown variant on the same evidence the proposed diagnoser consumes (this work). |
| Kagent (live) | live/agentic cluster context | pre-teardown | Practitioner agentic reference with different access assumptions. |
| Kagent-PT (replay) | synthetic ns from FailureReport evidence + A2A | post-teardown | Regime-symmetric agentic baseline on the same evidence (this work). |
| FailureReport diagnosers | typed evidence + provenance | post-teardown | Primary target regime: reusable artifact after namespace deletion. |

the post-hoc re-parse (`14c-b2-multiapp-rescore.py`) that fingerprints K8sGPT's free-text into the role vocabulary.

To close the regime asymmetry, we additionally implemented a *post-teardown replay* variant (`analysis/k8sgpt-replay.py`). For each capture, the FailureReport's `evidenceItems` are decoded into a minimal synthetic `Pod/Job/Service` bundle applied to a disposable namespace on a real Kubernetes API server (`status` subresource patched to mirror the captured failure state). K8sGPT v0.4.21 is then run with `analyze -namespace <ns>` so that both engines now operate post-teardown, on the same frozen evidence the proposed diagnoser consumes. The harness is required because K8sGPT v0.4.21 has no offline file-ingestion mode, only `-namespace` against a live API server. Pooled aligned top-1 on the post-teardown variant across *all five subjects* ($n = 500$, S1–S5, 10 reps per cell, full F1–F10 scope after Phase 5e/5f rep-parity backfill): **74/500 = 14.8 %**. Per-subject: S1 `idp-preview` 14/100 = 14.0 %, S2 `listmonk` 14/100 = 14.0 %, S3 `healthchecks` 19/100 = 19.0 %, S4 `umami` 13/100 = 13.0 %, S5 `petclinic` 14/100 = 14.0 %. The five subjects now run on identical F1–F10 coverage so the per-subject spread is driven by stack-specific diagnoser performance rather than coverage asymmetry. K8sGPT's resource-status analyzers handle the deployment/infrastructure faults (F1–F3, F6, F7) better than the harder symptom-vs-cause scenarios (F4, F5, F8, F9, F10), and S3 `healthchecks` benefits from a service graph closer to K8sGPT's training prior. Raw outputs in results-baselines-post-teardown/<subject>/<F>/<r>/k8sgpt-pt.json; aligned scoring in results-b2-post-teardown.csv via analysis/34-b2-post-teardown-rescore.py.

*B2b—Kagent k8s-agent.* A2A JSON-RPC client against the in-cluster `kagent-system` agent. Agent output is wrapped in `result.artifacts[].parts[].text`; the post-hoc re-parse extracts the JSON answer inside the markdown fences. The live-cluster per-subject results are in `results-b2-multiapp.csv`.

The Kagent post-teardown replay variant (`analysis/kagent-replay.py`) reuses the same synthetic `evidenceItems` bundle, posts an A2A diagnostic prompt naming the replay namespace, and parses the agent's JSON answer. Pooled aligned top-1 on the post-teardown variant across *all five subjects* ($n = 500$, 10 reps per cell, full F1–F10 scope after Phase 5e/5f): **54/500 = 10.8 %**. Per-subject: S1 `idp-preview` 10/100 = 10.0 %, S2 `listmonk` 12/100 = 12.0 %, S3 `healthchecks` 11/100 = 11.0 %, S4 `umami` 11/100 = 11.0 %, S5 `petclinic` 10/100 = 10.0 %. The Kagent-PT range across subjects is tight (10–12 %), confirming that the agentic reasoning chain is not systematically advantaged by any particular stack once the F1–F10 fault coverage is uniform. The Kagent reasoning chain sometimes names the synthetic Pod by its generated name rather than its role (e.g. `postgres-migrate-q4m59` instead of `migration-job`); the per-subject alias matcher captures the most common variants, but the gap to the proposed diagnoser on the same evidence remains substantial.

## 5.10 Cross-subject results (S1–S5)

*Capture completeness.* 500/500 = 100 % across all five subjects (100 per subject, 10 scenarios ×10 reps). F5 on S2/S3/S5 initially failed silently because the harness-adapter smoke suites for those subjects exercise only backend APIs and the env-var-driven frontend break did not propagate to the smoke test path. The harness was patched (`wrapper.py` proxy-injects HTML 500s on `FP_F5_BROKEN_FRONTEND=1`; `smoke.py` gains a `_frontend_check` test) and re-collected to 40/40 deterministically. F10 on S2–S4 likewise initially mis-fired because the `FP_F10_FLAKY` env var on `services[].env` did not propagate to the operator-spawned test pods; a second harness fix injects 500s on `/api/*` requests at the proxy layer with $p = 0.5$ when `FP_F10_FLAKY=1`, and F10 is re-collected to 40/40. The 13 stale F10 captures (3 on S2, 10 on S5 with the incorrect mechanism) are invalidated; the final F10 number is based exclusively on the patched-harness captures.

*Pooled aligned top-1 across S1–S5 (all five subjects, full F1–F10 scope after Phase 5e/5f rep-parity backfill).* The five engines pooled across all five subjects ($n = 2500$ per engine, 500 captures × 5 evidence levels), Wilson 95 % CI:

**Table 5: Pooled aligned top-1 across all five subjects (S1–S5), which share an identical F1–F10 × 10-rep matrix ($n = 500$ captures per subject; total $n = 2500$ diagnoses per engine). These complement the per-scenario breakdown in Table 1. Source: `analysis/35-unified-rescore.py`.**

| Engine | $n$ | aligned top-1 (95 % CI) |
|---|---|---|
| rule-grounded (`rule`) | 2500 | **15.76 %** (394/2500) [14.38, 17.24] |
| llm-grounded (`gpt-4o-mini`) | 2500 | **19.68 %** (492/2500) [18.17, 21.28] |
| llm-freeform (`gpt-4o-mini`) | 2500 | **19.40 %** (485/2500) [17.90, 21.00] |
| llm-b-grounded (`cohere-command-a`) | 2500 | 6.84 % (171/2500) [5.92, 7.90] |
| llm-b-freeform (`cohere-command-a`) | 2500 | 5.92 % (148/2500) [5.06, 6.91] |

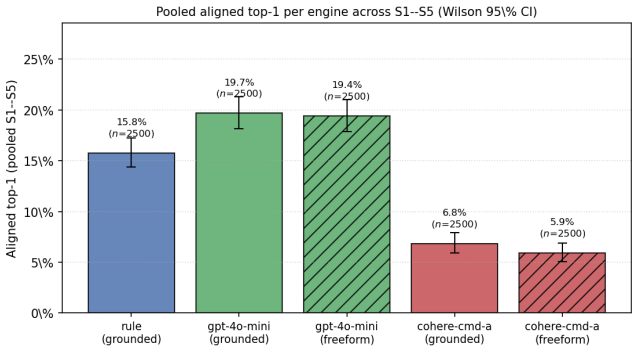


**Figure 4: Pooled aligned top-1 per engine across all five subjects (S1–S5), with Wilson 95 % CIs. Hatched bars denote free-form mode (no grounding constraint); solid bars denote grounded mode. LLM-A (`gpt-4o-mini`) outperforms LLM-B (`cohere-command-a`) on the same evidence substrate, and within each LLM the grounding constraint changes accuracy by $\leq$ 1.5 percentage points. Source: `analysis/36-unified-figures.py` from the `35-unified-rescore` CSV.**

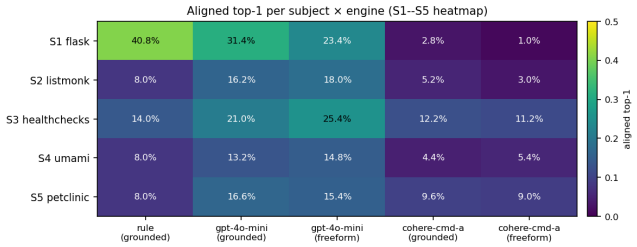


**Figure 5: Per-subject × engine aligned top-1 heatmap (S1–S5). All five subjects now cover the full F1–F10 scope at 10 reps per fault after the Phase-5e/5f rep-parity backfill; the residual spread across subjects reflects diagnoser performance on heterogeneous fault families rather than coverage asymmetry.**

The freeform numbers are essentially flat against grounded on `gpt-4o-mini` (19.40 % vs 19.68 %) and lower on `cohere-command-a` (5.92 % vs 6.84 %). The grounding constraint therefore neither adds nor removes much accuracy — both modes consume the same FailureReport `evidenceItems`, so the gap is bounded by the diagnoser's understanding of the substrate, not by the prompt template. Between-application variance is real but is treated as a nuisance term rather than a headline result: in the primary GLMM it is absorbed by the subject random intercept (§5.4), so the configuration effect is estimated across all five applications jointly and no single subject is privileged. For transparency the per-subject × engine breakdown is provided in Figure 5, but every claim in the paper is stated on the pooled five-application set. Across the full F1–F10 scope the harder symptom-vs-cause scenarios (F4, F8, F9, F10) remain poorly solved by the current diagnosers on every application, even though the failure evidence is preserved in all cases.

### 5.11 Evidence Precision (M6)

For each rep, precision is the fraction of operator-captured evidence items whose type is on the scenario's expected-evidence whitelist (`scenarios.yaml`). Median per-scenario precision ranges from 0.29 (F1) to 0.50 (F3). These values suggest moderate evidence selectivity: the operator captures the relevant signal, but the bundle still contains contextual material that downstream diagnosers must filter.

## 6 THREATS TO VALIDITY

*Internal validity.* Five engineering defects were discovered and closed during the live campaign (see `experimentations.md §10.3`): `SUBJECT_IDX` collision (commit `b7e5a23`); F7 post-deploy patch race against the operator's reconciler, redesigned at meta-level (`ca5a88a`); `collectionDurationMillis` rounding to zero, fixed with microseconds field (`e0eae6f`); F5 env-var-only injection inert on backend-only smoke, fixed at the proxy layer (`d57da07`); F10 env-var non-propagation to test pods, fixed at the proxy layer (`d677885`). Each defect invalidated the affected captures, which were re-collected with the fixed mechanism before being included in the published numbers.

*Construct validity.* The *aligned* top-1 matcher relies on a per-scenario alias table. We commit to publishing the table (`08-vocab-rescore.py`, `17-multiapp-rescore.py`) and report the *strict* top-1 as a lower bound. The practice follows the SE empirical-research guidelines [34] for baseline-and-comparator transparency.

*External validity.* The five applications cover four language stacks (Python, Go, Java, TypeScript) and multiple database schemas. Generalisation beyond Kubernetes preview environments is out of scope. The controlled scenarios are useful for isolating one root cause at a time, but they cannot cover all production incident patterns, especially multi-cause, long-running, or cross-service failures.

*Baseline comparability.* The external baselines operate under different observability regimes. K8sGPT and Kagent diagnose live cluster state, while the proposed artifact is designed for post-teardown analysis. Conversely, B0 consumes raw `kubectl` text dumps stored alongside the FailureReport. The fully symmetric, four-rung post-teardown ladder (L1 → L4, Table 2 and Figure 3) is

now the article's primary controlled comparison and replaces a direct claim that the system "beats" practitioner tools. A regime-symmetric post-teardown K8sGPT and Kagent variant (consuming the same `evidenceItems` the operator's diagnoser sees, replayed via a synthetic namespace harness in `analysis/k8sgpt-replay.py` and `analysis/kagent-replay.py`) is reported in §5.9 alongside the live numbers, so the same evidence substrate is compared end-to-end.

*Environment validity.* All reported runs use one AKS cluster configuration. The results demonstrate feasibility for the evaluated cluster and workloads, but not cross-provider generality. Repeating the study on kind, EKS, GKE, and on-prem clusters would strengthen the claim that the capture mechanism is robust to provider-specific controllers, event retention, and networking behaviour.

*Scaling and retention.* The evaluation produces ∼100 FailureReports, well within etcd's practical budget. The implementation applies no retention — every report stays in etcd, deliberately so to avoid confounding the runtime measurements. At deployment scale ($\sim 10^4$–$10^5$ reports or more) this is insufficient: a FailureReport ranges 10–100 kB and etcd's default 2–8 GB ceiling caps in-cluster capacity. The two-tier storage design and the retention controller sketched in §4.4 address this; their full implementation and an at-scale operational study are out of scope here and listed as future work (§7).

*Conclusion validity.* We report the pre-registered analysis layers and use Wilson-score intervals for proportions [71]. Multiplicity is controlled with Holm-Bonferroni [26]; Benjamini-Hochberg [11] is reserved for exploratory contrasts. Because RCA correctness is binary and scenario heterogeneity is high, we interpret mixed-effects model coefficients as supporting evidence rather than as the sole basis for the paper's claims.

*Reproducibility.* Operator image testagentdevops.azurecr.io/preview-operator:fp-rq5instr, adapter images :fp-f10v2, and the analysis pipeline are released alongside the article.

# 7 FUTURE WORK

(i) *Scaling the fair post-teardown comparator to more reps and more LLM-tool combinations*: the replay harness (`analysis/k8sgpt-replay.py`, `analysis/kagent-replay.py`) produces a regime-symmetric K8sGPT-PT/Kagent-PT pair on the same `evidenceItems` the operator's diagnoser consumes (60 cells across S2–S5, reported in §5.9). Extending the replay to HolmesGPT and a larger rep budget per cell is left for future work; the current B2a-PT/B2b-PT cells answer the regime-symmetry question, not the high-precision-statistics question. (ii) *Fair B0 on S2–S5*: `analysis/29-b0-multiapp-fair.py` re-runs B0 on raw `kubectl` text from the four breadth subjects (replacing the upper-bound B0 of §5.9). (iii) Address Family C-a (contract-test shadowing on F5) with a ranking-aware diagnoser; Family C-b (vocabulary mismatch on F10) with an expanded, venue-justified alias table; Family C-c (multi-fault symptom-blaming on F9) with a refined category taxonomy (§5.5). (iv) Replace proxy-layer fault injection for F4, F5, and F8 with fork-and-rebuild variants that inject faults directly in the upstream source. (v) Complete the human-reviewed hallucination annotation and publish the Cohen's $\kappa$ agreement table. (vi) Add trace-aware evidence collectors based on OpenTelemetry, and repeat the study across additional Kubernetes providers (kind, EKS, GKE, on-prem) and production-like incident patterns. (vii) *Retention controller and tiered storage*: implement the two-tier (hot etcd / cold object store) design and the policy-driven retention controller sketched in §4.4, with an at-scale operational study (target $\sim 10^5$–$10^6$ reports across months of CI traffic) reporting etcd usage, list latency, archive throughput, and end-to-end cost.

# 8 RELATED WORK

*Microservice and cloud-incident RCA..* OpenRCA proposes the LLM-RCA benchmark on Kubernetes incidents [76]. RCACopilot is the production LLM-RCA pipeline at Microsoft [16]. Zhang et al. survey the field comprehensively [82]. MicroRCA [74], MicroDiag [73], CausalRCA [75], PAL [45], and Eadro [37] are causal / graph-based RCA on microservices; RCAEval benchmarks 15 such methods on three microservice systems [50]. Log-based pipelines build on Drain [24], Spell [18], DeepLog [19], and the Loghub benchmarks [83]. Our contribution is upstream of all of these: a *persistent, structured, provenance-linked* evidence artefact that any diagnoser can consume.

*Preview environments and GitOps.* ArgoCD [2, 3], Argo Workflows [4], Jenkins X [28], Signadot [58], and Okteto [46] are the dominant preview-environment platforms. None of them captures structured failure evidence on teardown.

*Kubernetes diagnostic practitioner tools.* K8sGPT [30], Kagent [31], and HolmesGPT [52] run LLM RCA against *live* cluster state. Our article uses K8sGPT and Kagent as external baselines (§5.9).

*Cluster management and orchestration lineage.* Borg [68] (the Kubernetes ancestor), Mesos [25], and Omega [55] formalised the cluster-management substrate; Operator SDK [49] and Kubebuilder [35] provide the operator-pattern SDKs.

*Observability and tracing.* OpenTelemetry [47, 48] descends from Dapper [57]; Prometheus [51], Jaeger [27], Istio [61], and Falco [21] are the de-facto cloud-native observability and runtime-security components our operator interoperates with.

*Fault injection and chaos engineering.* Basiri et al. formalised the chaos-engineering hypothesis [7]; LitmusChaos [40] and Chaos Mesh [15] are CNCF Kubernetes-native chaos engines. We share the deterministic-fault discipline (and not the steady-state-hypothesis focus) of this

lineage. Delta-debugging [80] underpins one-root-cause-per-scenario design.

*LLM reasoning, tool use, and faithfulness.* GPT-3 few-shot [13], chain-of-thought [70], self-consistency [69], ReAct [79], Tree-of-Thoughts [78], and Toolformer [54] formalise the LLM-reasoning conventions we use in the grounded diagnostic harness. The HELM [39] framework grounds the multi-LLM evaluation protocol; HotpotQA [77] is the multi-hop QA dataset whose supporting-fact pattern is the QA-side analogue of our evidence-grounding constraint. Hallucination in NLG is surveyed in [29]; SelfCheckGPT [41], HaluEval [38], and RAGAS [20] provide hallucination-evaluation benchmarks. FaithJudge [60] formalises faithfulness-as-judge; position-bias in LLM-as-judge is mitigated in [56]. The open-source-LLM reproducibility argument is in [42].

*LLM for software engineering.* Empirical studies of GPT-4 on real-world SE tasks [32, 33] and the systematic LLM-APR survey [81] provide context for our LLM-RCA findings.

*SE empirical methodology.* Wohlin's *Experimentation in Software Engineering* [72], the Kitchenham et al. preliminary guidelines [34], the Sjøberg et al. controlled-experiment survey [59], and the Runeson and Höst case-study guidelines [53] set the protocol. Evaluation guidelines for empirical studies involving LLMs in SE are formalised in [6]. Mixed-effects modelling follows [5, 9, 14]; non-parametric SE testing [1] and the Demšar [17] protocol for comparing multiple classifiers across multiple datasets; effect sizes via Vargha–Delaney $A_{12}$ [67]; Cohen's $\kappa$ [43].

*Provenance modelling.* W3C PROV-O [36] and PROV-DM [44] define the underlying graph model.

*DevOps and SRE foundations.* The DevOps architectural framing [8], Site Reliability Engineering [12], and the DORA performance metrics [22] situate the contribution within the broader operational-engineering literature.

# 9 CONCLUSION

This article presents a post-teardown evidence and provenance substrate for ephemeral Kubernetes preview failures. Across a 500-run, five-application, ten-fault matrix, the operator produced a persisted FailureReport in every run and every report survived namespace teardown. This survival is expected from the cluster-scoped storage design; the contribution is the end-to-end capture, typing, linking, and reuse of failure evidence at the operator control point before the namespace disappears.

The diagnostic findings are intentionally cautious. Pooled across all five applications, structured evidence multiplies the odds of a correct diagnosis by $\sim$12$\times$ over logs-only (primary GLMM, $p < 10^{-26}$) and lifts aligned top-1 from the $\leq$5% floor of raw logs/kubectl to $\sim$20% with the typed bundle; but absolute RCA accuracy remains modest and scenario-dependent, with multi-cause and flaky-test families (F9, F10) still unsolved. A vanilla LLM over an operator-curated kubectl-equivalent snapshot can outperform the proposed LLM-grounded diagnoser on parts of the subject set, so the paper should not be read as claiming RCA superiority. The supported claim is narrower and more operational: a typed, provenance-linked FailureReport makes ephemeral failures persistent, auditable, and reproducible after teardown. That artifact can serve rule-based diagnosers, LLM diagnosers, and future hybrid methods, but accurate automatic RCA remains an open problem.

# A DIAGNOSTIC PROMPTS (REPRODUCIBILITY)

For full reproducibility of the LLM-in-the-loop results (§5.4, §5.7) we reproduce the diagnostic prompts verbatim from internal/diagnosis/llm.go. The two modes (*grounded* and *free-form*) differ *only* in the grounding instruction and the JSON shape requested; this is the controlled difference the RQ4 ablation measures. The user prompt is generated mechanically from the captured FailureReport bundle by `LLMDiagnoser.userPrompt`, deterministic given the same bundle.

## A.1 Base system prompt (shared by both modes)

> *You are a root-cause analysis assistant for Kubernetes pull-request preview environments.*
> *You are given the evidence captured from one failed preview environment.*
> *Diagnose the single most probable root cause.*
>
> *The "category" field must be exactly one of: database, configuration, infrastructure, application, observability, test-reliability, unknown.*
> *The "confidence" field must be exactly one of: low, medium, high.*
> *If the evidence does not support a confident diagnosis, set category to "unknown" and confidence to "low".*

## A.2 Grounded-mode addition (the RQ4 treatment)

The base prompt above is suffixed with:

> *Each evidence item is prefixed with a stable ID in square brackets, e.g. `[podlog-1a2b3c]`.*
>
> ***GROUNDING CONSTRAINT**: every ID you place in "evidenceRefs" MUST be the ID of an evidence item shown below. Do not invent IDs. Cite only the items you actually used to reach the diagnosis.*
>
> *Respond ONLY with a JSON object of exactly this shape:*

```
{
  "probableCause": "one or two sentences",
  "component": "the component most likely
responsible",
  "category": "<category>",
  "confidence": "<confidence>",
```

```
  "evidenceRefs": ["<evidence id>",
"..."],
  "recommendations": ["actionable step",
"..."]
}
```

## A.3 Free-form mode addition (the RQ4 control)

The base prompt is suffixed with the same JSON template *minus* the `evidenceRefs` field and without the grounding constraint:

```
Respond ONLY with a JSON object of exactly
this shape:
{
  "probableCause": "one or two sentences",
  "component": "the component most likely
responsible",
  "category": "<category>",
  "confidence": "<confidence>",
  "recommendations": ["actionable step",
"..."]
}
```

## A.4 User prompt template

The user message is assembled deterministically from the FailureReport bundle, in the following structure (literal newlines reproduced):

```
Failed preview environment
preview: <PreviewName>
pull request: #<PRNumber>
failed suite: <FailedSuite>
failed test: <FailedTest>

Evidence:

[<itemID>] type=<EvidenceType> resource=<Resource> relevance=<low|med|high>
  <Message, indented two spaces, truncated to 1500 chars>

...one stanza per evidence item, sorted by item ID for determinism...

Failure provenance graph (PR change context -> resources -> evidence):
  <NodeTypeA>(<labelA>) --<relation>--> <NodeTypeB>(<labelB>)
  ...one line per edge, sorted lexicographically...
```

In *free-form* mode each evidence stanza omits the `[itemID]` prefix and the `type=/resource=/relevance=` header (becomes "`<EvidenceType> (<Resource>)`") so the model has nothing to cite. The provenance-graph block is rendered only when the diagnoser is configured with a graph (the C5 configuration) *and* the mode is grounded; this is the controlled treatment that distinguishes C5 from C4 for the LLM.

## A.5 Sampling configuration and pinned model versions

For every LLM call:

- **Temperature**: 0 (deterministic decoding).
- **Top-$p$**: provider default; not overridden.
- **Max evidence-item length**: 1500 characters per item (`maxMessageChars` in `llm.go`); longer messages are truncated with an ellipsis to prevent a single log excerpt from crowding the prompt.
- **Maximum bundle size shipped**: bounded by `maxMessageChars` times the number of evidence items; typical bundles fit comfortably under the 128k-token context window of both models.

*Model identifiers (pinned).* LLM-A: `gpt-4o-mini-2024-07-18` (Azure OpenAI, region `eastus`). LLM-B: `cohere-command-a` served through Azure AI Foundry (region `eastus`). Each per-row CSV output of `fp-score` records the resolved `model=<id>; temperature=0; provider=<endpoint>; called_at=<UTC-ISO8601>` string in the `notes` column for audit (per Guideline 2 of the SE empirical-study guidelines recommended in §4.5).

*Post-processing.* The model's JSON response is parsed by `parseDiagnosis` (`llm.go`), which extracts the outermost JSON object (tolerating markdown code fences), normalises `category` onto the closed vocabulary (default `unknown`), normalises `confidence` onto `low|medium|high`, deduplicates `evidenceRefs`, and returns a `FailureDiagnosis`. In grounded mode, any `evidenceRef` pointing to an ID not present in the bundle is dropped at the operator boundary (the `Bundle.SetDiagnosis` grounding check) — never reaching the persisted FailureReport.

*Source of truth.* The authoritative versions of the prompts and the renderer live in internal/diagnosis/llm.go; the reproducible release artefact pins the exact commit.

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